

Health Economics -- Assignment 5.

Part I – MC (2 points each)

Question 1. Network and referrals

A patient is enrolled in a **PPO** plan through her employer. She wants to see an in-network dermatologist for a rash.

Which statement best describes how her plan works in this situation?

- A. She must get a referral from a primary care physician or the visit will not be covered
- B. She can see the in-network dermatologist without a referral, paying the usual in-network cost-sharing
- C. She can see the dermatologist only if the dermatologist is out-of-network
- D. The PPO covers specialty care only after the deductible is met, with no difference between in- and out-of-network providers

Question 2. HDHP and cost-sharing

A patient is in an HDHP with:

- Annual deductible: 1,500 dollars
- Co-insurance after deductible: 20 percent
- Out-of-pocket maximum: 4,000 dollars

In one year she incurs 5,000 dollars in covered, in-network medical expenses.

How much does she pay out of pocket?

- A. 1,500 dollars
- B. 2,200 dollars
- C. 4,000 dollars
- D. 5,000 dollars

Question 3. Ex ante vs ex post moral hazard

Which of the following is the clearest example of **ex post** moral hazard in health insurance?

A person with generous insurance drives more recklessly because they know medical bills will be covered

B. A person with full coverage for physical therapy schedules extra sessions mainly for mild soreness

C. A young adult decides not to get vaccinated because they underestimate their risk of infection

D. A person quits their gym membership after losing employer-sponsored insurance

Question 4. Expected utility and buying insurance

A risk-averse individual faces a 5 percent chance of incurring 10,000 dollars in medical costs next year and a 95 percent chance of incurring 0 dollars. An insurer offers **full coverage** for a premium of 600 dollars.

Which of the following best explains why this person might rationally buy the insurance?

- A. The premium is lower than the expected loss, so the insurance is actuarially fair
- B. The premium equals the expected loss, so there is no administrative load
- C. The premium is higher than the expected loss, but they are willing to pay extra to avoid financial risk
- D. The premium is higher than the expected loss and risk-averse people always reject such contracts

Question 5. Moral hazard in physical therapy use

A patient with full insurance has a 0 dollar copay for physical therapy (PT). The market price per PT session is 80 dollars. Their demand for PT is given by

$Q = 12 - 0.1P$, where Q is the number of sessions and P is the patient's out-of-pocket price per session.

Which of the following best describes the direction of change in PT use and spending when the patient moves from full insurance (0 copay) to being uninsured (pays full price)?

- A. Quantity demanded rises, and patient spending rises
- B. Quantity demanded rises, and patient spending falls
- C. Quantity demanded falls, but total social spending on PT is unchanged
- D. Quantity demanded falls, and patient spending per session rises

Bonus Question: In 1-2 sentences, explain why. (2 points)

Part II – Problems

Problem 1. Adverse Selection in an insurance market (10 points)

A small non-profit health insurer is considering offering a **single community-rated plan** to 500 potential enrollees. Each person knows their own expected annual health care costs, but the insurer can only observe age and a few basic characteristics, not each individual's true risk.

There are two risk types in the population:

- 250 **low-risk** individuals each expect to spend **400 dollars** per year
- 250 **high-risk** individuals each expect to spend **1,200 dollars** per year

Assume people will **buy the insurance if the premium is less than or equal to their own expected costs**.

a. If **everyone enrolled**, what would be the **average expected cost per person**, and what would be the **actuarially fair community premium** for the plan? Show the simple calculation. (2 points)

b. Suppose the insurer sets the premium equal to the average cost you calculated in part a. (3 points)

- Will **low-risk** individuals be willing to enroll at that premium?
- Will **high-risk** individuals be willing to enroll?

Explain briefly for each type.

c. If low-risk individuals decide **not to enroll**, leaving only high-risk individuals in the pool, what is the **new average expected cost** among enrollees? If the insurer adjusts the premium to equal this new average cost, what happens to the premium relative to your answer in part a? (2 points)

d. In 2–3 sentences, explain how this example illustrates **adverse selection** and why it can lead to a “death spiral” in a community-rated insurance market. Use the lecture language about “sicker, high-cost individuals” and “healthy people dropping out as premiums rise.” (3 points)

Problem 2. Expected loss, fair premium, and risk aversion (10 points)

A 30-year-old individual faces the following risk next year: with probability 2 percent, they will incur medical expenses of 50,000 dollars; with probability 98 percent, they will incur 0 dollars in medical expenses. A health insurer offers a full-coverage policy (no cost-sharing) for one year.

a. Compute this person’s **expected medical spending** next year. (2 points)

b. What is the **actuarially fair premium** for full insurance, ignoring administrative costs and profit? (2 points)

c. Suppose this risk-averse individual is willing to pay up to **1,500 dollars** for full insurance. What is the **maximum dollar administrative load** the insurer can charge and still sell the policy to this consumer? Explain briefly. (3 points)

d. In 1–2 sentences, explain the relationship between **expected loss**, **administrative load**, and **willingness to pay** in this example, and how risk aversion justifies an above-fair premium. (3 points)